FR695867.1:10012-20758
uncultured Desulfobacterium sp.
NAD_Gly3P_dh_N NAD_Gly3P_dh_C UPF0758_N RadC ? PGK VTT_dom cpYpsA DUF6794 ? HigB-like_toxin HTH_3 Y1_Tnp Bac_DnaA_C ? ? ? DUF772 DDE_Tnp_1 Zn_ribbon_8
LN848257.1:480967-498935
Thiocapsa sp. KS1
? ? AI-2E_transport Patatin ? MmeI_N MmeI_hel MmeI_Mtase MmeI_TRD MmeI_C HTH_3 HigB-like_toxin HTH_24 AsnC_trans_reg LeuA_dimer IPMS_D2 HMGL-like Helicase_C SNF2-rel_dom DUF389 DUF3040 ? HTH_1 LysR_substrate
DABN01000043.1:71679-79431
Candidatus Gastranaerophilales bacterium HUM_16
HTH_17 Phage_integrase Phage_int_SAM_5 ? ? ? ? HTH_3 HigB-like_toxin SLFN_AlbA_2 HsdM_N
GG693852.1:46338-59905
Leptospirillum ferrodiazotrophum
? ? ? Secretin ? ? HTH_3 HigB-like_toxin Phage_pRha T2SSE Hom_end_hint T2SSE FumaraseC_C SLT ? ? Caspase ANKs ANKs
DMAS01000066.1:1201-5611
Anaerolineae bacterium
RF-1 PrmC_N Methyltransf_31 Sua5_yciO_yrdC Sua5_yciO_yrdC HATPase_c_3 HSP90 HSP90 DUF7010 HigB-like_toxin HTH_3
DATAKL010000057.1:190533-210455
Ignavibacteriales bacterium
Rotamase_2 SurA_N Rotamase_3 Rotamase AAA_3 AAA_lid_2 AI-2E_transport PTA_PTB ? L6_membrane HigB-like_toxin HTH_3 GlgE_dom_N_S Alpha-amylase GLGE_C Alpha-amylase Malt_amylase_C Mak_N_cap APH Glyco_hydro_77 Ribonuc_red_2_N Ribonuc_red_lgC Ribonuc_red_lgC TSCPD ? Ala_racemase_N
DRPK01000020.1:192-1767
Persephonella sp.
HigB-like_toxin HTH_3 MoeA_N
DASCNB010000166.1:1-8014
Nitrospinota bacterium
? HTH_3 HigB-like_toxin ? DUF2130 NucS_C ? FSH1 Hydrolase_4 LbH_EIF2B NTP_transferase 23S_rRNA_IVP
DATFBA010000196.1:29147-9173
Geomonas sp.
AAA_lid_14 HTH_3 HigB-like_toxin Pyr_redox_dim Pyr_redox_2 VTT_dom E1_dh Transketolase_C Transket_pyr 2-oxoacid_dh E3_binding Biotin_lipoyl Rhomboid DUF7575 PAS
DATEUF010000365.1:3354-17289
Pyrinomonadaceae bacterium
Glyco_hydro_47 Acetylase tRNA-synt_1e DALR_2 CysS_C AIRS AIRS_C Aminotran_1_2 MerR_1 HTH_3 HigB-like_toxin OsmC Ligase_CoA ATP-grasp_2 MauE RMMBL Beta-Casp Lactamase_B_2 Ligase_CoA CoA_binding NDK DUF4440
DAOLDS010000446.1:1-4992
Anaerolineae bacterium
? HigB-like_toxin HTH_3 PMT_2
DAORVD010000229.1:7505-8565
Caldilinea sp.
CYTH GyrI-like ? HTH_3 HigB-like_toxin ? AbiEii AbiEi_4 ? ?
DAOMOK010000005.1:286412-309407
Chloroflexota bacterium
MS_channel_1st_1 MS_channel_1st_1 MS_channel_1st_1 MS_channel_1st_1 FHA PD40 Peptidase_S9 Receiver Glutaredoxin Pyr_redox_2 DUF1565 Beta_helix DUF11 DUF11 PKD_4 PKD PKD_4 PKD_4 PKD_4 PKD PKD_4 DUF1565 NosD PKD PKD PKD PKD PKD ? HerA-FtsK HAS-barrel HigB-like_toxin HTH_3 Glyco_hydro_10 LVIVD LVIVD Acetylase TPRs TPRs AP-ATPase HTH_31 DUF5128 NHL NHL NHL NHL Zn_ribbon_8
DATHXR010000571.1:86-2544
Arenicellales bacterium
HigB-like_toxin HTH_3 HipA_C Couple_hipA ?
DATHSF010000251.1:2498-7499
Thioalkalivibrio sp.
RHH_5 BrnT_toxin ? Aldo_ket_red DUF6064 ? HTH_3 HigB-like_toxin DUF2442 Ribonuc_red_sm
DATIPE010000311.1:199-4764
Thermoanaerobaculia bacterium
HTH_3 HigB-like_toxin CSD CSD ? DUF2071 Cu-oxidase Cu-oxidase_3
DASZRZ010000478.1:1-11885
Terriglobales bacterium
OTCace OTCace_N Arginosuc_syn_C Arginosuc_synth ASL_C2 Lyase_1 HTH_3 HigB-like_toxin Peptidase_M50 Bac_DNA_binding DCD SLBB_2 SLBB_2 Poly_export MerR_1 DnaJ DnaJ_C HSP70
DATMDU010000006.1:141720-17568
Pyrinomonadaceae bacterium
Mrr_cat ? MethyltransfD12 Phage_integrase ? ? HigB-like_toxin HTH_3 CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg DUF1579 DUF4256 EAL GGDEF PAS EAL GGDEF PAS Peptidase_M16 Peptidase_M16 Peptidase_M16_C
DATMAT010000023.1:338408-350634
Thermodesulfobacteriota bacterium
DUF2845 FTP Peptidase_M4 Peptidase_M4_C Lipase_GDSL_2 FAD_binding_3 TLC MFS_1 Aldo_ket_red HigB-like_toxin HTH_3 CopG_antitoxin BrnT_toxin OsmC WbqC LTXXQ ?
DATLZH010000604.1:878-6395
Blastocatellia bacterium
CHRD HTH_3 HigB_toxin HigB-like_toxin HTH_3 Glyco_transf_9 PEMT DUF3108 Hemerythrin
DASQOO010000187.1:105-18122
Steroidobacteraceae bacterium
? Sulfatase_C Sulfatase Sulfatase ? AbiEi_4 AbiEi_1 AbiEii WYL HTH_37 HigB-like_toxin HTH_3 DUF4143 Peptidase_S8 Reg_prop Reg_prop Reg_prop Reg_prop Reg_prop Reg_prop Y_Y_Y His-Kinase His-Kinase Receiver Receiver Hpt Receiver GGDEF EAL HMBD E1-E2_ATPase Hydrolase
DASRFE010000182.1:22860-31595
Burkholderiales bacterium
? GST_N_3 GST_C DUF1272 HicB_lk_antitox Glyoxalase ? HTH_3 HigB-like_toxin Unstab_antitox HicB_lk_antitox HicA_toxin ? DUF4926 DDE_Tnp_1 DUF4440
DASRSL010000089.1:94927-14220
Roseiarcus sp.
Glucodextran_N Glyco_hydro_15 Glyco_hydro_15 CBM_20 AcnX_swivel_put AcnX HigB-like_toxin HTH_3 NrtR_WHD NadA FAD_binding_2 QRPTase_N QRPTase_C PTA_PTB Malic_M malic WXXGXW
WPIJ01000018.1:33847-68225
Rhodocyclaceae bacterium
FAD_binding_5 CO_deh_flav_C Fer2 Fer2_3 Fer2_2 Ald_Xan_dh_C MoCoBD_1 MoCoBD_2 CoA_transf_3 Peripla_BP_6 TPP_enzyme_C POR DUF6537 DUF4347 Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll FG-GAP FG-GAP FG-GAP FG-GAP FG-GAP FG-GAP FG-GAP Beta-Roll HigB-like_toxin HTH_3 NnrS AMP-binding AMP-binding_C AAA_2 ClpB_D2-small Usp Usp Peptidase_C11 He_PIG PPC He_PIG PPC PPC PPC PPC PPC PPC PPC PPC PPC PPC PPC PPC Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll
LADQ01000045.1:1-7905
Hoeflea sp. BRH_c9
Transket_pyr Transketolase_C Biotin_lipoyl E3_binding 2-oxoacid_dh DUF308 DUF308 Lipase_GDSL_2 DUF465 Transgly_assoc LIAS_N Radical_SAM HigB-like_toxin ? Polyketide_cyc
LKUC01000085.1:239-2126
Smithella sp. SDB
Adhesin_E ? HTH_3 HigB-like_toxin ?
LGFV01000125.1:122-2918
Actinobacteria bacterium 66_15
? HigB-like_toxin HTH_3 ? VanZ
JACDAY010000345.1:1-8554
Blastocatellia bacterium
Aminotran_3 HigB-like_toxin HTH_3 Thioredox_DsbH Beta-AFase-like_GH127_cat ? GAF_2 GGDEF STY-kinase TPRs TPRs TPRs
JACNIG010000128.1:377-6144
Candidatus Desulfatibia vada
HTH_3 HigB-like_toxin HTH_3 N_methyl GspH N_methyl N_methyl PilX_N Vfa1 ParE_toxin
JADBLR010000741.1:1-2221
Planctomycetota bacterium
Mur_ligase_M Mur_ligase HigB-like_toxin HTH_3 Mur_ligase_C Mur_ligase_M
WGMY01000039.1:38826-50462
Alphaproteobacteria bacterium
DUF2497 PCMT fn3_5 TauD TauD ? HTH_3 HigB-like_toxin ? ? DUF6804 ? ? SpoVG ?
JACOTC010000264.1:1-3688
Nitrospirae bacterium
Lactamase_B RNase_J_b_CASP RMMBL HTH_3 HigB-like_toxin Ribonuc_red_lgC Ribonuc_red_lgC Ribonuc_red_lgN
JACQHF010000002.1:265-9757
Candidatus Kerfeldbacteria bacterium
? Mur_ligase_M Mur_ligase_C HigB-like_toxin HTH_3 ? ? ? ? ? RNA_pol_Rpb2_2 RNA_pol_Rpb2_3 RNA_pol_Rpb2_45 RNA_pol_Rpb2_6 RNA_pol_Rpb2_7
JACQQS010000057.1:1-13316
Candidatus Omnitrophota bacterium
TPM_phosphatase ApbA_C ApbA ? Trans_reg_C Receiver His-Kinase His-Kinase HAMP HTH_3 HigB-like_toxin ? DUF7387 HicA_toxin DUF2283 MgtE_N CBS CBS MgtE
JACQYM010000061.1:25616-10112
Deltaproteobacteria bacterium
Peptidase_S26 HigB-like_toxin HTH_3 UPF0158 DUF4143 MetallophosC Calcineurin MetallophosN HDOD CutA1 TPRs TPRs TPRs TPRs TPRs
JACRMI010000128.1:112-8171
Chloroflexi bacterium
OrfB_IS605 ? Prok-JAB ? ThiF HigB-like_toxin HTH_3 DNA_pol3_beta DNA_pol3_beta_2 DNA_pol3_beta_3 ? ParB_N HTH_ParB ? ? ?
JACRDE010000269.1:839-3376
Desulfomonile tiedjei
DNA_binding_1 ? ? Acyl_transf_3 HigB-like_toxin HTH_3
JADJBM010000030.1:13928-25385
Betaproteobacteria bacterium
HAD_SAK_2 ? ? ? DUF4258 ARG ? HigB-like_toxin HTH_3 FAD-oxidase_C FAD_binding_4 TRAM_2 Radical_SAM UPF0004 PHB_acc PHB_acc PHB_acc_N adh_short adh_short Abhydrolase_1 PhaC_N
JADJFV010000001.1:127875-139937
Leptospiraceae bacterium
UbiA COX15-CtaA COX15-CtaA ANKs ANKs MaoC_dehydratas FAS1_DH_region DEAD Helicase_C RecQ_Zn_bind HTH_3 HigB-like_toxin ? ? DUF6364 7TMR-DISMED2 7TMR-DISM_7TM Guanylate_cyc Uma2 ? Fer4_12
JADJOJ010000010.1:483490-500082
Candidatus Leptofilum gracile
N6_N4_Mtase N6_N4_Mtase N_BRCA1_IG N_BRCA1_IG Band_7_1 HTH_17 ? HTH_DeoR A2L_zn_ribbon DUF2207 DUF2207_C HTH_3 HigB-like_toxin SBP_bac_3 ? ? GDP_Man_Dehyd FtsJ S4 His-Kinase His-Kinase TPR_MalT TPRs
VGNN01000042.1:3944-6077
Chloroflexi bacterium
Ribonuc_red_lgC Ribonuc_red_lgC HigB-like_toxin HTH_3 CAP Enolase_C Enolase_N UPF0150 GYD LDcluster4 Hydrolase_4 Abhydrolase_1
VHCZ01000177.1:3076-12572
Verrucomicrobia bacterium
? Cupredoxin_1 CarboxypepD_reg DUF218 Flagellin_C Phage_integrase ? HTH_3 HigB-like_toxin ? AAA_25 ? ?
VGSK01000123.1:2216-7549
Deltaproteobacteria bacterium
Amidohydro_1 HigB-like_toxin HTH_3 ? HepT-like_2 Polbeta Ldh_2 SdrD_B SdrD_B SdrD_B Cupredoxin_1 Glycos_transf_2
JAFGJY010000114.1:26370-33239
Candidatus Omnitrophota bacterium
DEAD Helicase_C B12-binding Radical_SAM DUF1318 ? ? Methyltransf_25 HTH_3 HigB-like_toxin
JAGACZ010000200.1:44-2657
Eggerthellaceae bacterium
HupF_HypC HypD HigB-like_toxin HTH_3 Cyt_bd_oxida_I
JAGBVH010000022.1:12199-26500
Alphaproteobacteria bacterium
PQ-loop ? RE_endonuc Helicase_C ResIII HTH_34 N6_N4_Mtase ? ? HigB-like_toxin ? ? ? ? Arginase ? ?
JAEPPY010000014.1:1-4627
Pseudomonadales bacterium
DUF4262 HTH_3 HigB-like_toxin ? bPH_4 ? BrnT_toxin DUF4440 HxlR APH
JAGODU010000330.1:301-3417
Prolixibacteraceae bacterium
Fic HTH_3 HigB-like_toxin PIN PhdYeFM_antitox ?
JAFQXR010000038.1:3235-12504
Eggerthellaceae bacterium
Zn_ribbon_IS1595 Nif11 DUF6350 HTH_3 HTH_3 HigB-like_toxin Zn_Ribbon_1 PIN HTH_3 HigB-like_toxin ? ? URO-D EamA EamA Rce1-like PIN ?
JAFRKV010000044.1:1893-5514
Lachnospiraceae bacterium
? HTH_3 HigB-like_toxin ? ? ? ? Acetyltransf_10 ? ABC_tran OB_MalK TOBE_2
JABICZ010000467.1:7409-15714
Deltaproteobacteria bacterium
NMO MeaB MM_CoA_mutase B12-binding ECH_1 ? HTH_3 HigB-like_toxin SnoaL_4 AMP-binding AMP-binding_C_3
JABHLS010000094.1:1-4291
Bacteroidetes bacterium
HigB-like_toxin DrHyd PALP DUF6377 Fic
JAHJBZ010000021.1:43189-52330
Patescibacteria group bacterium
Dus PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C Palm_thioest DUF2878 HEAT_PBS ? AAA_33 HTH_3 HigB-like_toxin DUF998 PAD_porph
JAFDKH010000296.1:1-4842
Deltaproteobacteria bacterium
? TSP_3 TSP_3 TSP_3 TSP_3 TSP_3 ? HigB-like_toxin HTH_3 ? ?
JAHWLB010000517.1:1-1950
Acidobacteria bacterium
AIRS_C AIRS HTH_3 HigB-like_toxin
JAIBBN010000017.1:45559-53812
Leptospirales bacterium
DDE_Tnp_IS66_C DDE_Tnp_IS66 ? ? ? ? DUF772 HTH_3 HigB-like_toxin DDE_Tnp_1_6 DUF772 SMI1_KNR4 LRR_EndoS
JAIVJD010000431.1:1045-5938
Acidobacteria bacterium
? HigB-like_toxin HTH_3 ? NAD_binding_4 Sterile AMP-binding AMP-binding_C Acyltransferase Epimerase
JAGQKN010000635.1:1-2636
Candidatus Omnitrophota bacterium
Dockerin_1 HigB-like_toxin HTH_3 AhpC-TSA GHMP_kinases_N GHMP_kinases_C
JAGRFM010000040.1:16580-22159
Nitrosomonas sp.
ATP-synt_C ATP-synt_A ATP-synt_I ABC_membrane ABC_tran Rotamase_3 His_Phos_1 HTH_3 HigB-like_toxin ?
JACKEV010000014.1:12407-27233
Alphaproteobacteria bacterium
? TPRs TPRs TPRs HEATs LTD Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE ? PEGA ? HigB-like_toxin HTH_3 ? FecR Sigma70_r4_2 Sigma70_r2 ? BolA TMEM164
JAFIEQ010000031.1:5900-25083
Nitriliruptoraceae bacterium
Peptidase_S15 YcaO YcaO Peptidase_S9 MatE MatE Methyltransf_25 HTH_3 HigB-like_toxin ? PG_binding_1 ABC_tran FtsX ? Whib ?
JAIPEN010000008.1:84108-9487
Desulfarculaceae bacterium
HigB-like_toxin HTH_3 ? AEP-TOTE ResIII Helicase_C ? HepT-like NTP_transf_2 Receiver His-Kinase His-Kinase PAS PAS PAS PAS PAS
JADFTE010000062.1:673-6780
Candidatus Neomarinimicrobiota bacterium
HigB-like_toxin HTH_3 ? Receiver NtrC-AAA RsmI_C TP_methylase HEATs HEATs HEATs HEATs BPD_transp_1
JADFMC010000036.1:3435-10700
Candidatus Neomarinimicrobiota bacterium
GDPD Penicil_amidase Thioredoxin Sigma70_r4_2 Sigma70_r2 zf-HC2 OsmC HTH_3 HigB-like_toxin Peptidase_M3 TOP_N
JALHTW010000159.1:2696-9816
Candidatus Aminicenantes bacterium
DUF4097 ChAPs TPRs ? Peptidase_M28 AbiEii ? HTH_3 HigB-like_toxin Type_ISP_C
WRKW01000005.1:26335-38353
Alphaproteobacteria bacterium
PseudoU_synth_2 ? NT5C GIY-YIG ? SpoVG ? HigB-like_toxin DNA_methylase RE_NgoFVII_C ? ? Resolvase Recombinase Zn_ribbon_recom ABC2_membrane_6 ?
JAHDVJ010000876.1:161-2631
Bryobacteraceae bacterium
PIN MazE_antitoxin HTH_3 HigB-like_toxin ? Sirtuin
JAMLHW010000003.1:285980-299872
Anaerolineae bacterium
DrrA1-3_C ABC_tran ABC2_membrane His-Kinase HisKA_3 GerE Receiver DUF6414 DUF6414 HTH_3 HigB-like_toxin Methyltransf_11 TPP_enzyme_C PFO_beta_C POR POR_N GntR FCD Hist_deacetyl LiaF-TM ABC_tran ABC_membrane
JAAENE010004421.1:1-4406
bacterium
HEATs HTH_3 HigB-like_toxin RloB ABC-ATPase
JAAEOE010000874.1:3633-3633
Proteobacteria bacterium
PemK_toxin ? HigB-like_toxin HTH_3 HTH_3
JAAEOG010001277.1:1-2252
bacterium
? HigB-like_toxin DDE_Tnp_1 DUF772 Peptidase_S8
JAAETZ010000203.1:1-3243
Chloroflexota bacterium
? HigB-like_toxin HTH_3 RHS_repeat ? ? ? ? ?
JANLIF010000225.1:1-5155
Candidatus Scalindua sp.
HigB-like_toxin HTH_3 DUF4143 AAA_14 ? HigB-like_toxin HTH_3 DUF3427 Helicase_C
JAILEK010000041.1:1-2222
Erysipelotrichaceae bacterium
YdjC MazG-like HTH_3 HigB-like_toxin ? ?
JAPOYZ010000476.1:42-3064
Candidatus Aminicenantes bacterium
HTH_3 HigB-like_toxin ? OmdA ?
JAPUIL010000481.1:1166-8122
Verrucomicrobia bacterium
? DHC FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 HTH_3 HigB-like_toxin ? AP_endonuc_2 Y1_Tnp ? AbiEii ? YafQ_toxin
JALLOK010000255.1:2321-4013
Acidobacteria bacterium AH-259-D05
UPF0175 ? ? HTH_3 HigB-like_toxin GAF HD HD_5
JAOMND010000084.1:60-8661
Oligoflexia bacterium
PfkB YicC-like_N YicC-like_C Guanylate_kin PDZ_6 ? STY-kinase SET HigB-like_toxin HTH_3
JAQUJK010000225.1:1-3613
Desulfoplanes sp.
PGM_PMM_I ? HEPN ? HTH_3 HigB-like_toxin Y1_Tnp Y1_Tnp
JAPPGS010000040.1:1-4030
Gemmatimonadota bacterium
Sulfatase HTH_3 HigB-like_toxin PhyH PhyH PhyH
JAOUMB010000078.1:1383-8484
Nitrospirota bacterium
? UPF0547 Helicase_C ResIII DUF6511 PDDEXK_1 AAA_25 ? HTH_3 HigB-like_toxin ?
JASJEI010000004.1:33523-52459
Pleurocapsa sp. MO_226.B13
? Biotin_lipoyl_2 HlyD_D23 Glyco_hydro_57 Aminotran_1_2 ? ? ? HigB-like_toxin HTH_3 BFN_dom ? ? DUF3479 CobN-Mg_chel ? TPS
SZPG01000073.1:15161-30002
Chloroflexi bacterium CFX6
Alpha-amy_C_pro Alpha-amylase Abhydrolase_1 Abhydrolase_6 F420H2_quin_red Aminotran_1_2 Malic_M THF_DHG_CYH_C malic ACT_4 Hydrolase adh_short_C2 HigB-like_toxin HTH_3 TauE ? HhH-GPD AA_permease_2 Rce1-like NDH2_C Pyr_redox_2
JAUXWS010000009.1:4080-15675
bacterium
DDE_Tnp_1 DUF4096 HTH_3 HigB-like_toxin Na_H_Exchanger TrkA_N TrkA_C DUF190 CRCB RecD_N AAA_30 UvrD_C_2 UvrD-helicase UvrD-helicase UvrD_C PDDEXK_1 Exonuc_V_gamma RecC_C STAS Sulfate_transp
JAVLBT010000227.1:186-1243
Gracilimonas sp.
CopG_antitoxin HTH_3 HigB-like_toxin
JAVVEY010000023.1:3684-17990
Longimicrobiales bacterium
Fic_N DUF1403 ? ? PNPase Fic Fic_N HTH_3 HigB-like_toxin ? Lactamase_B_2 ? HD_5 HD GAF PFK Peptidase_S9 DPPIV_N ?
JAWVYP010000058.1:5956-12260
Akkermansiaceae bacterium
Radical_SAM Gly-zipper_YMGG Gly-zipper_YMGG Thioredoxin_7 DsbD_TM DsbD_N YHYH ? HTH_3 HigB-like_toxin
JAXHMB010000004.1:639451-653212
Balneolaceae bacterium
Peptidase_S58 HSP33 ELFV_dehydrog ELFV_dehydrog ELFV_dehydrog_N DHH DHHA1 RecJ_OB HTH_3 Couple_hipA HipA_C HigB-like_toxin HTH_3 ? DUF676 Type_ISP_C Type_ISP_C Eco57I N6_Mtase
JAXHNK010000007.1:282754-293996
Anaerolineales bacterium
? Patatin Catalase Catalase-rel HTH_3 HigB-like_toxin Transposase_20 DEDD_Tnp_IS110 HTH_3 HigB-like_toxin ? Pyr_excise ? ? PemK_toxin RHH_1 PIN
JAYFNV010000027.1:43621-57728
Anaerolineaceae bacterium
Thioredoxin_3 ArsP_1 LMWPc Nitroreductase Nitroreductase HI0933_like OmdA DUF1801 HTH_3 HigB-like_toxin TiaS_TCKD ? Receiver His-Kinase His-Kinase Sigma70_r4 Receiver SBP_bac_8 BPD_transp_1 BPD_transp_1
JAZBNR010000018.1:18885-34564
Erythrobacter sp.
Transglutaminase Bact_transglu_N Alpha-E CP_ATPgrasp_2 Abhydrolase_1 DUF3617 DUF3617 ResIII UvrB_inter Helicase_C UvrB UVR HigB-like_toxin ? Peptidase_S10 ? Thioredoxin_7 DsbD_N FAD_binding_3 ? HEM4
VXNI01000134.1:3975-7688
Chloroflexi bacterium
DrrA1-3_C ABC_tran ABC2_membrane HigB-like_toxin HTH_3 HTH_3 HigB-like_toxin GFO_IDH_MocA_C3 DEAD Helicase_C LRR_14 DUF7507
JAACAH010000030.1:25485-40653
candidate division KSB1 bacterium
GMC_oxred_N GMC_oxred_C GFO_IDH_MocA GFO_IDH_MocA_C3 ? ? MGH1-like_GH Glyco_hydro_63 Yip1 CoA_binding Ligase_CoA HigB-like_toxin HTH_3 MazE_antitoxin PIN Asp_protease_2 FtsX FtsX MacB_PCD FtsX MacB_PCD FtsX MacB_PCD
JAACAH010000767.1:3111-10292
candidate division KSB1 bacterium
? Fe-S_biosyn NifU B12-binding Radical_SAM Radical_SAM Mob_synth_C HTH_3 HigB-like_toxin adh_short ? UPF0102 RNase_HII MlaC OEP OEP MlaC DUF3047
JAABYG010000010.1:159-2612
Nitrosopumilaceae archaeon
HigB-like_toxin HTH_3 ? Trypan_PARP Somatomedin_B
JAAURV010000018.1:3521-19981
Hyphomicrobiales bacterium
ParE_toxin Pyr_redox_2 Fer4_20 AbiEii ? SGL ILVD_EDD_N ILV_EDD_C Aldose_epim HigB-like_toxin HTH_3 ? PriA_C Helicase_C Zn_ribbon_PriA DEAD PriA_3primeBD Anth_synt_I_N Chorismate_bind Aminotran_4 TAL_FSA ABM
JAAUSB010000250.1:206-3830
Rhodospirillaceae bacterium
Glyoxalase_6 MORN ? HTH_3 HigB-like_toxin HTS
JAAZPA010000239.1:1219-11709
Syntrophorhabdus sp.
N6_N4_Mtase HTH_3 HigB-like_toxin SBP_bac_5 BPD_transp_1 BPD_transp_1_N Virul_fac_BrkB 5_3_exonuc_N 5_3_exonuc DNA_pol_A NirdL-like_HTH AsnC_trans_reg2 zf-C2H2_ZNF142_18 TraB_PrgY_gumN
JAAZPP010000256.1:63-5572
Deltaproteobacteria bacterium
? WCX HigB-like_toxin HTH_3 UPF0047 ? DUF429 ATPase_gene1 ? HTH_8 AAA_lid_14 NtrC-AAA Receiver
WSZU01000099.1:28-4631
Anaerolineales bacterium
Rieske UCR_Fe-S_N Cytochrom_B_C Cytochrome_B HigB-like_toxin HTH_3 MFS_1 ? HTH_5
JABRXF010000003.1:479726-491798
Hongsoonwoonella zoysiae
GTP_cyclohydro2 DHBP_synthase HTH_26 ? Methyltransf_25 TRAM TRM5-TYW2_MTfase tRNA_U5-meth_tr S4 FtsJ ? HTH_3 HigB-like_toxin His_Phos_1 adh_short_C2 FAD-oxidase_C FAD_binding_4 SRAP ? NUDIX DUF2385
JABUFB010000018.1:43329-60633
Armatimonadetes bacterium
? ? CarboxypepD_reg ADH_zinc_N CBM_4_9 DUF4838 AP_endonuc_2 Methyltransf_25 HigB-like_toxin HTH_3 ? GFO_IDH_MocA_C3 GFO_IDH_MocA Glyco_hydro_43 Hepar_II_III_C GATase CPSase_sm_chain Amidohydro_1
MEQN01000032.1:10362-6764
Betaproteobacteria bacterium RBG_16_58_11
HigB-like_toxin HTH_3 DUF6364 PIN Rieske Ring_hydroxyl_A EamA EamA SpoIIAA-like GAF GGDEF
MWCZ01000087.1:5020-11425
Candidatus Latescibacteria bacterium ADurb.Bin168
Calci_bind_CcbP ? UvrD_C AAA_19 DUF7309 DUF6930 Zn_ribbon_Top1 NERD HTH_11 WYL WCX HTH_3 HigB-like_toxin
NBLZ01000090.1:1-7778
Anaerolineaceae bacterium 4572_5.1
DEAD Helicase_C WHD_HRQ1 HNH MZB N6_N4_Mtase Eco57I Endonuc-BsobI HigB-like_toxin HTH_3 RNase_H_2 Spermine_synth
PFOM01000012.1:18988-11569
Candidatus Peregrinibacteria bacterium CG_4_10_14_0_2_um_filter_38_24
zf_Tbcl_2 ? Phage_integrase Phage_int_SAM_4 HTH_3 HigB-like_toxin DUF1297 DUF1246 DUF1297 DUF1246 RelB Hexapep LbH_EIF2B NTP_transferase Glyco_transf_7C Glycos_transf_2 Aminotran_1_2 Transket_pyr Transketolase_C
QZJB01000087.1:5453-13938
Candidatus Parcubacteria bacterium
TIR ? ? MUG113 ? ? HigB-like_toxin ? ? Peptidase_M78 ? ? Multi_ubiq Prok-E2_E HTH_3
QZIQ01000059.1:6084-15743
Desulfobacteraceae bacterium
Lactamase_B_2 DUF4345 DJ-1_PfpI Abhydrolase_1 TetR_N Phage_int_SAM_4 ? PDZ_6 TPRs TPRs HigB-like_toxin HTH_3 HTH_3 TetR_N Abhydrolase_6 ABC2_membrane ABC2_membrane_7 ABC_tran adh_short_C2 ?
QNBP01000071.1:3956-5681
Spirochaetes bacterium
? ? DUF362 Fer4_5 Fer4_5 Fer4_7 Fer4_7 Fer4 ? ? Phage_integrase HigB-like_toxin
QMLL01000100.1:1-6447
Deltaproteobacteria bacterium
PemK_toxin HTH_3 HigB-like_toxin ? OmpA TraL TraE TraK_C TraK_N TrbI TrbC
RFKB01001722.1:1-1130
Caldilineae bacterium
adh_short_C2 HigB-like_toxin
RPOY01000935.1:204-3391
Acidobacteria bacterium
DUF7660 DUF7129 ? ? ? HTH_3 HigB-like_toxin
QNZC01000078.1:6326-8320
Persephonella sp.
DUF4198 ? Phenol_MetA_deg CbiM ? ABC_tran HigB-like_toxin HTH_3 HepT-like Polbeta SLFN_AlbA_2 Acylphosphatase zf-HYPF zf-HYPF Sua5_yciO_yrdC
RDXB01000040.1:49479-70884
Oscillatoriales cyanobacterium
Competence DUF4131 GAF PAS ANAPC4_WD40 Aminotran_1_2 ? HTH_3 HigB-like_toxin DUF2442 NNH2 NACHT HEATs HEATs HEATs DUF533 HEATs Beta-Roll Beta-Roll Beta-Roll Peptidase_M10 Fe-ADH Peptidase_M20 E1-E2_ATPase Hydrolase
RDXC01000475.1:25289-35315
Oscillatoriales cyanobacterium
Acyl-CoA_dh_1 Acyl-CoA_dh_N Cyclase DUF1995 BPD_transp_1 Methyltransf_11 HTH_3 HigB-like_toxin MoCF_biosynth Psb28 DUF1565 Trypan_PARP SLH SLH ?
RECN01000100.1:120435-132032
Spirochaetaceae bacterium
SBP_bac_1 GntR FCD PhdYeFM_antitox PIN HTH_3 HigB-like_toxin Sulfatase SGSH_C His-Kinase His_kinase HAMP dCache_1 Receiver HTH_18 ? Sulfatase DUF3450
CP070698.1:562484-574094
bacterium
PEP-utilizers ? FtsK_4TM FtsK_alpha FtsK_SpoIIIE FtsK_gamma UPF0149 SEC-C PIN PhdYeFM_antitox HigB-like_toxin HTH_3 RsmF-B_ferredox NusB DUF116 Formyl_trans_C Formyl_trans_N PriA_C RsgA_GTPase BacA Pep_deformylase
CP129490.1:182597-197751
Melioribacteraceae bacterium
Thiolase_C Thiolase_N 4HBT 3HCDH 3HCDH 3HCDH_N PALP PaaA_PaaC PaaB PaaA_PaaC HigB-like_toxin HTH_3 FeS_assembly_P Zn_ribbon_PaaD Peptidase_M1_N Peptidase_M1 PD40 PD40 PD40 ? HerA-FtsK
NZ_AFWV01000001.1:273188-291996
Thiocapsa marina 5811
? Trypsin Methyltransf_25 GerE DUF2126 Transglutaminase Bact_transglu_N ? HigB-like_toxin QueF FGGY_C FGGY_N PAS PAS GGDEF EAL PDH_N PDH_C Aminotran_1_2 CM_2 PDT ACT 2-Hacid_dh_C DUF4438_C DUF4438
NZ_JAWXXV010000001.1:3330193-3343122
Sphingomonas echinoides
Fumerase Fumerase_C GIY-YIG Lactamase_B_2 ? ? DUF3617 ResIII UvrB_inter Helicase_C UvrB UVR HigB-like_toxin PRC Sugar_tr Sugar_tr ? ? Amidase DUF2721 ?
NZ_FNLL01000032.1:1-4301
Desulfobacula phenolica
PhdYeFM_antitox ParE_toxin ? HTH_3 ? HTH_3 TIR HigB-like_toxin HigB-like_toxin
NC_019682.1:1527014-1540296
Calothrix sp. PCC 7507
? DnaA_N-like_STI DNA_pol3_gamma3 DNAX_ATPase_lid DNA_pol3_delta2 Intein_splicing DNA_pol3_delta2 zf-C2HC_2 ParE_toxin Glyco_tranf_2_3 HTH_3 HigB-like_toxin ? DRTGG AAA_26 DNA_processg_A MAPEG DUF520 GST_N GST_C_3 ? RHH_1
NZ_CAAAIQ010000053.1:569-2219
Legionella waltersii
HTH_3 IstB_IS21 HigB-like_toxin
NZ_APIZ01000361.1:55-875
Dolichospermum circinale AWQC310F
HigB-like_toxin ? ?
NZ_JQKJ01000019.1:90891-101004
Desulfobacter vibrioformis DSM 8776
TMP-TENI HATPase_c_4 DUF364 Thioredoxin Zn_ribbon_Thio2 PhdYeFM_antitox HTH_3 HigB-like_toxin N_methyl N_methyl Unstab_antitox His-Kinase HisKA_3 HAMP dCache_2 Bactofilin Bactofilin PIN N_methyl DUF2442
NZ_LVJX01000010.1:220-12726
Candidatus Thiodiazotropha endoloripes
? RadC ? ? nSTAND_NTPase5 HTH_3 ? Trypsin ? HigB-like_toxin Astacin ? ? ? Phage_integrase
NZ_LT896716.1:60232-75232
Geobacter sp. DSM 9736
DHQS_C DHQ_synthase AP_endonuc_2 UbiA HTH_3 HigB-like_toxin B12-binding Radical_SAM SAICAR_synt SurA_N_3 Rotamase_2 Rotamase_2 MreB_Mbl Glycos_transf_2 SIS_2 Hydrolase_like Hydrolase_6 ? TatD_DNase Phosphodiest
NZ_FNRJ01000015.1:19681-37427
Marinobacterium iners DSM 11526
DUF411 AIM24 REase SWI2_SNF2 T1R_D3-like HTH_3 HigB-like_toxin GIY-YIG DUF4357 Methylase_S Methylase_S DNA_ligase_OB_2 DNA_ligase_A_M Phosphonate-bd EAL GGDEF HAMP Cytochrome_CBB3 ? DUF1853 EAL HDOD HsdM_N N6_Mtase
NZ_ROQN01000098.1:21150-32340
Leptospira barantonii
Peptidase_S26 ATPase_gene1 ? OSCP ATP-synt_ab_C ATP-synt_ab ATP-synt_ab_N ATP-synt_B ATP-synt_A HigB-like_toxin HTH_3 HpcH_HpaI TetR_N CorA LytTR Receiver ATP-synt_C ?
NZ_QLSV01000003.1:90656-115495
Flavobacterium lacus
CHU_C Ig_2 CHU_C SprB SprB SprB SprB SprB SprB SprB SprB SprB SprB SprB SprB SprB SprB SprB SprB SprB SprB DUF7619 FolB GldM_3rd GldM_4th HTH_3 HigB-like_toxin MacB_PCD FtsX NamZ_C NamZ_N Methyltransf_23 DUF5030 TonB_C DUF2059 Radical_SAM PorP_SprF PorP_SprF CxxCxxCC
NZ_JAIZBL010000008.1:17912-24350
Leptospira bandrabouensis
HTH_3 RelE HTH_3 Zn_Tnp_IS91 Y2_Tnp ? HigB-like_toxin Polbeta HepT-like Y2_Tnp
NZ_JAAKEG010000005.1:94805-112390
Adlercreutzia sp. ZJ304
Pterin_bind HPPK PHO4 PhoU_div Ribosomal_L2_N Ribosomal_L2_C Ribosomal_S19 Trigger_N FKBP_C Trigger_C Lactamase_B HigB-like_toxin HD_4 RelA_SpoT TGS RelA_AH_RIS ACT_4 DHH DHHA1 RecJ_OB DEAD Helicase_C BPL_LplA_LipB BPL_C Pribosyltran HTH_3 BioY FtsH_ext AAA AAA_lid_3 Peptidase_M41
NZ_BLRY01000231.1:924-830
Candidatus Hakubella thermalkaliphila
HigB-like_toxin DUF123
NZ_CAJHCF010000017.1:36332-46473
Gulosibacter hominis
DprA_WH HATPase_c_4 SLFN_AlbA_2 ? Acetyltransf_7 FitA-like_RHH PIN PhdYeFM_antitox YoeB_toxin HigB-like_toxin HTH_3 DUF6290 Peptidase_C15 ABM ABC_tran ABC_tran Acetylase MgtE CBS CBS MgtE_N
NZ_JAHREP010000005.1:143279-159327
Marinobacter ramblicola
DctM MS_C MS_TIM-barrel MSG_insertion MS_N HbpS-like FAD-oxidase_C FAD_binding_4 FAD_binding_4 CCG CCG Fer4_8 HTH_3 HigB-like_toxin Unstab_antitox HGLS choice_anch_I YceI Glycos_transf_4 Trans_reg_C Receiver His-Kinase His-Kinase HAMP
NZ_CP078069.1:2433641-2448262
Comamonas sp. NLF-1-9
Peptidase_C26 Ribosomal_S30AE SSB CorA AAA_16 UvrA_inter UvrA_DNA-bind ABC_tran ResIII HigB-like_toxin ? ? FAD_binding_3 FAA_hydrolase FAA_hydrolase_N Cyclase SGL MFS_1 HTH_3
NZ_JACNCX010000011.1:116520-128848
Rhizobium sp. C4
zf-HC2 Sigma70_r4_2 Sigma70_r2 Pyr_redox_2 Phage_integrase ? Bep_C_terminal T4SS-DNA_transf HigB-like_toxin ? T2SSE TrbI CagX VirB8 TrbL ? DUF736
NZ_JARANV010000007.1:123645-136578
Halomonas sp. 25-S5
FlgO HigB-like_toxin HTH_3 DUF3368 ? Flagellin_C Flagellin_N Glyco_transf_41 Glyco_transf_41 TPRs TPR_CcmH_CycH ? Peptidase_M78 HTH_3 HicB_lk_antitox ? ? OmpA MotB_plug TctC SLT_4
NZ_DBFXCT010000403.1:92-2011
Slackia isoflavoniconvertens
HigB-like_toxin HTH_3 ? ORF6N
NZ_JAUFPL010000027.1:14235-30337
Halomonas ramblicola
TetR_C_13 TetR_N MFS_1 GAF EAL ? YicC-like_C Nitroreductase HTH_3 HigB-like_toxin VMAP-M0 dCMP_cyt_deam_1 Na_sulph_symp Biotin_lipoyl E3_binding 2-oxoacid_dh Transketolase_N Transketolase_N PDH_E1_M Transketolase_C_1 Metallophos_2 Biotin_lipoyl Biotin_lipoyl Pyr_redox_2 Pyr_redox_dim
NZ_JANEIL010000354.1:1-3318
Thiolapillus sp.
CRISPR_assoc HigB-like_toxin HTH_3 Cas_Cas1 Cas_Cas5d Cas_CT1975
NZ_CAUEOF010000030.1:27727-8664
Bilophila wadsworthia
? Collar Topoisom_bac Toprim ? TrbM HigB-like_toxin HTH_3 TrbL ? ? ? ? TrbI
NZ_JAVDPC010000043.1:30682-37665
Cerasicoccus frondis
HK97-gp10_like Tail_completion_gp17 ? Phage_tail_T ? ? ? HigB-like_toxin HTH_3 Acetylase
NZ_OY781616.1:1013531-1029127
Maridesulfovibrio sp.
? MCPsignal ? ? ? TMEM43 HTH_3 LPD38 ANKs ANKs ANKs ANKs ANKs ANKs ANKs ANKs ? DUF5675 ? Gp49 HigB-like_toxin LPD38
NZ_JAVEPN010000018.1:32232-47950
Trichlorobacter sp.
TP_methylase RsmI_C UDPG_MGDP_dh_N UDPG_MGDP_dh UDPG_MGDP_dh_C GDP_Man_Dehyd DivIC ? Arg_tRNA_synt_N tRNA-synt_1d DALR_1 SPOR HigB-like_toxin NrfD Fer4_11 Fer4_7 Molydop_binding Molybdopterin Molybdop_Fe4S4 TAT_signal ? MobB HTH_3
NZ_JAYGHQ010000011.1:16412-31644
Crocosphaera sp. XPORK-15E
FA_desaturase MacB_PCD FtsX Biotin_lipoyl_2 HlyD_D23 HlyD_2 NQR2_RnfD_RnfE XPB_DRD ResIII Helicase_C HTH_3 HigB-like_toxin tRNA-synt_2d Phe_tRNA-synt_N AtpR ATP-synt_A ATP-synt_C ATP-synt_B ATP-synt_ab_C ATP-synt_ab ATP-synt_ab_N Transferase TOBE_2 ABC_tran